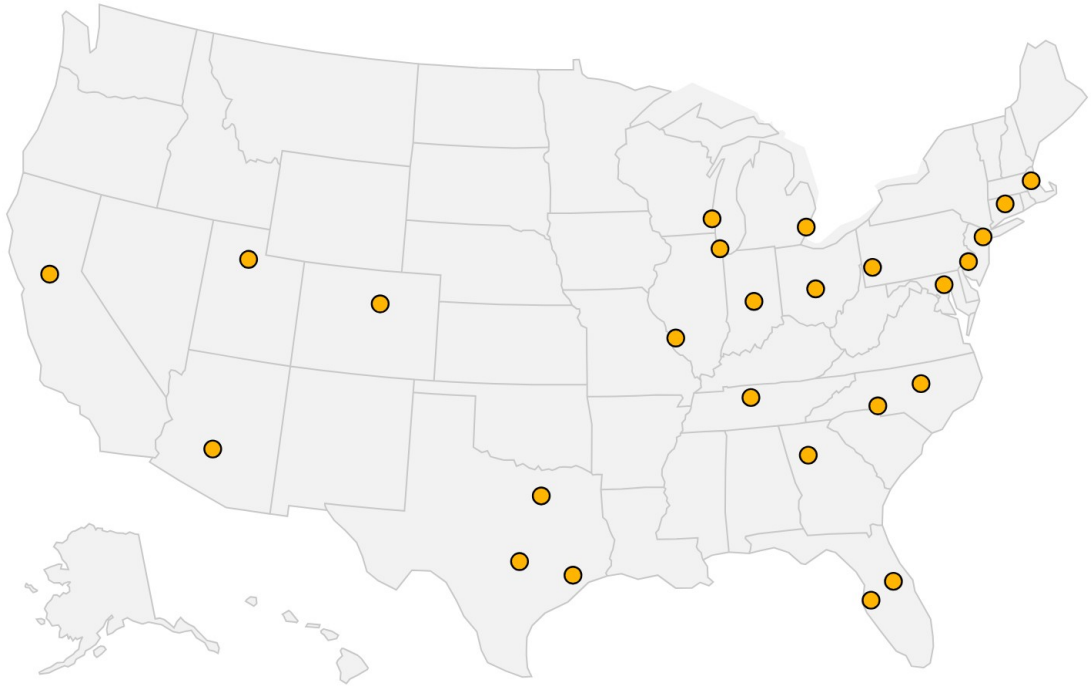
Stockflow

Branch-level demand forecasting and inventory rebalancing

Right product at the right branch, so surgeons, hospitals and their patients are not left waiting.

Bryce Biyeba
Data Science Intern · October 2026



25 branches · 300 SKUs · 4,889 branch × SKU combinations

62% of inventory dollars sit above target while 411 branch positions face stock-outs

5.8%

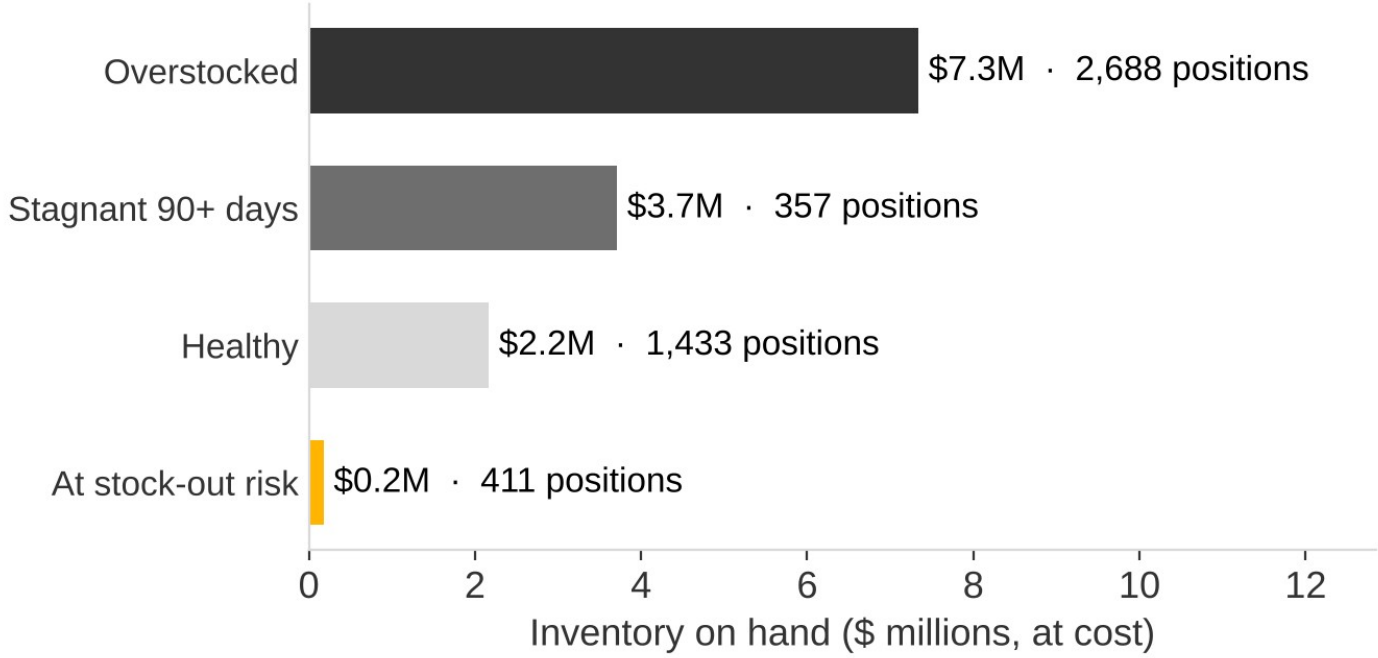
of branch × SKU days were stock-out days (3 years)

$8.3M

above target, of $13.4M on hand

$3.7M

stagnant: no sale in 90+ days



Inventory on hand by position status, at cost

One pipeline turns three years of branch data into transfers a planner approves



sales, stock moves, POs, promotions

outages, returns, stock-out days

LightGBM + statistical, P10 / P50 / P90

safety stock, reorder point, risk

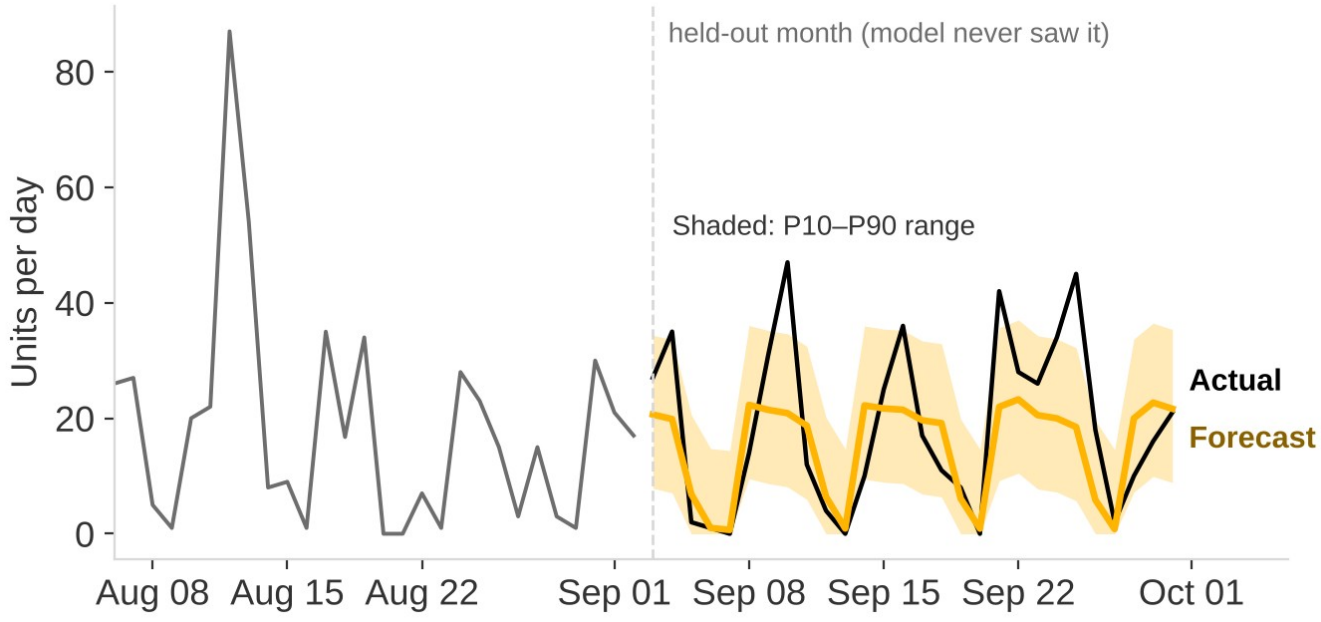
min-cost transfers under constraints

plain-language reason per move

nothing moves without sign-off

Every step checks its inputs; every assumption is documented in the repository.

Forecasts are 34% more accurate than the seasonal baseline on a month the model never saw

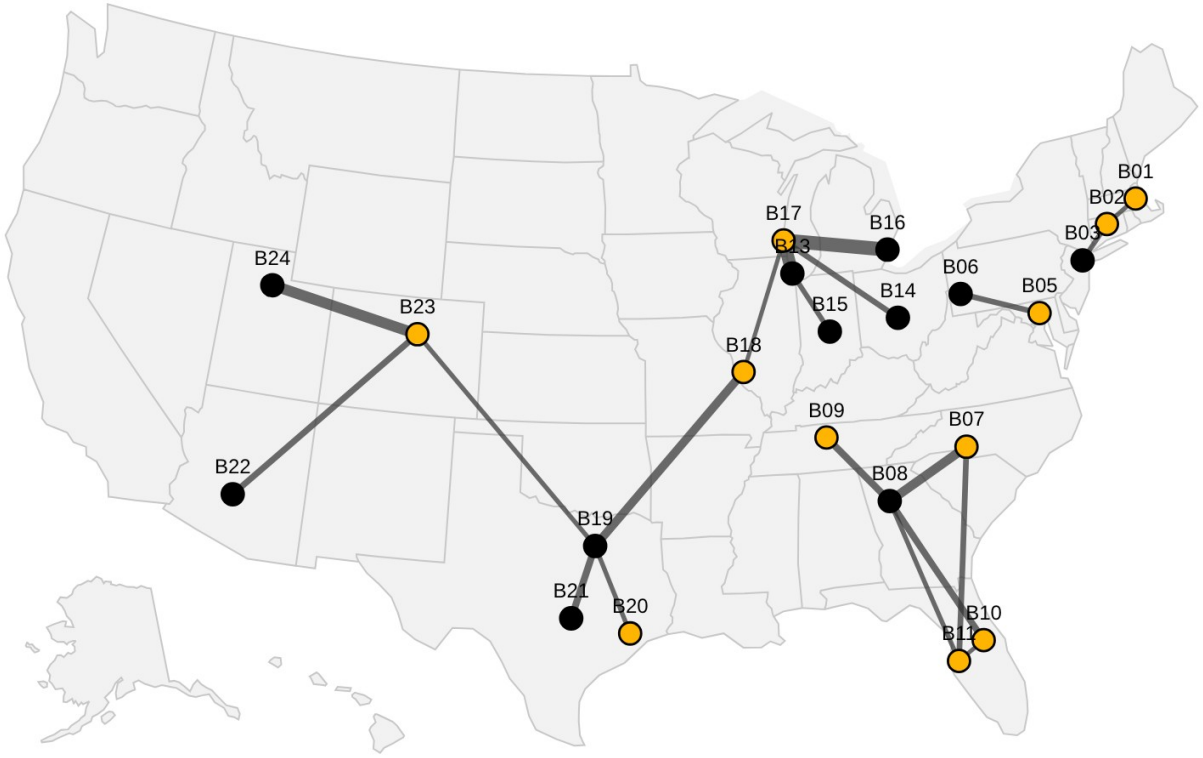


Branch B24, SKU S0057 (Tools), 2026-09-03 to 2026-09-30

Model (held-out month)	WAPE	MAE	Bias
Seasonal naive (baseline)	74.9%	3.38	+1.2%
Exponential smoothing	56.6%	2.55	+2.5%
Croston (intermittent)	57.3%	2.58	+0.9%
LightGBM (global ML)	49.7%	2.24	-0.7%
Selected per segment	**49.8%**	**2.24**	**-0.8%**

WAPE = absolute error ÷ actual units; MAE in units per day. P10–P90 band held 84% of actual days (target 80%).

883 transfers cost $45k to move and save $203k net, $10k more than a rule of thumb



Top 20 of 265 corridors · line width = units · black = sending branch, gold = receiving

From → To	SKU	Units	Net savings	Reason
B10 → B17	S0190	192	$3,957	Excess → shortfall
B19 → B23	S0190	202	$3,899	Excess → shortfall
B14 → B19	S0285	180	$3,231	Excess → shortfall
B10 → B07	S0190	122	$3,050	Excess → shortfall
B16 → B17	S0134	185	$3,015	Excess → shortfall

Each move ships with a plain-language reason a branch manager can check.

Rebalancing frees $1.1M of excess and avoids 84% of expected lost sales

$1.1M

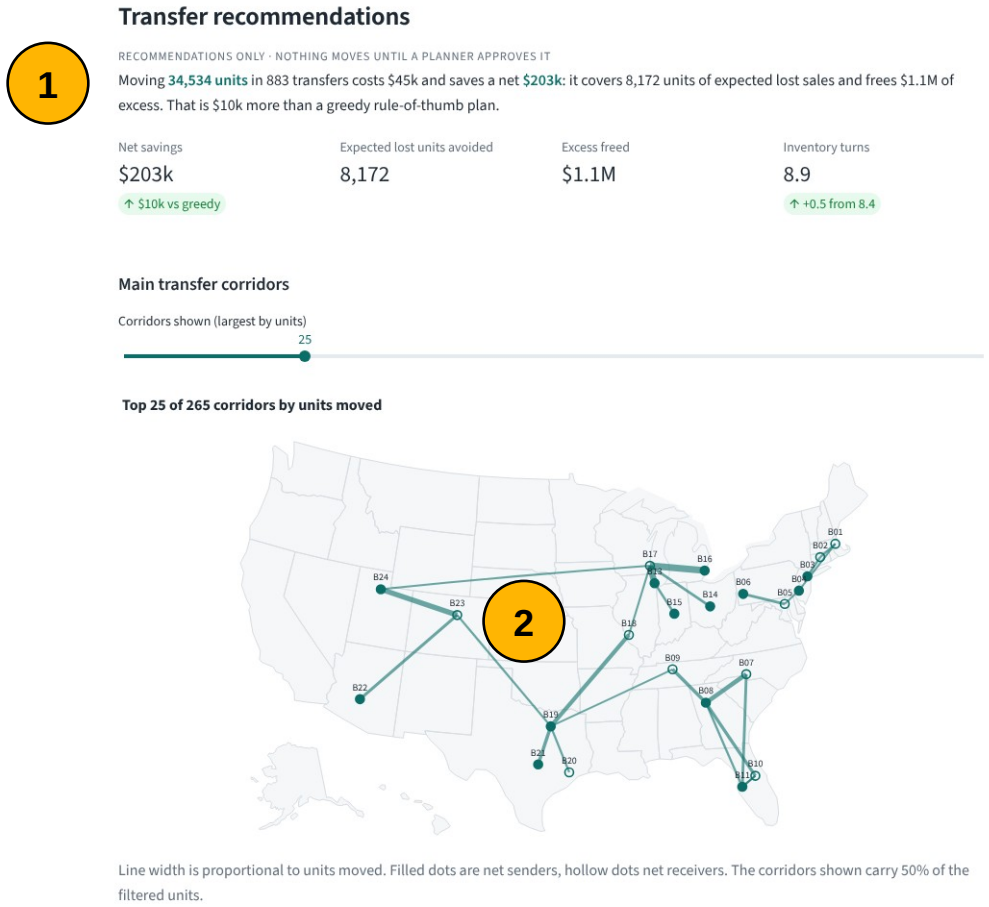
excess inventory put back to work

8,172 lost-sale units avoided

Measure	Before	After
Excess above target	$8.3M	$7.2M
Annual carrying cost of excess	$2.18M	$1.92M
Expected lost-sale units (lead time)	9,725	1,552
Positions at stock-out risk	411	72
Inventory turns	8.4	8.9
Net savings (one cycle)	–	$203k

Model estimates for one planning cycle versus doing nothing.

A branch manager sees what to move, why, and approves it on one screen



3

From	To	SKU	Decide	Note	Status	Reason
B10	B17	S0190			Pending	Excess to shortfall
B19	B23	S0190			Pending	Excess to shortfall
B14	B19	S0285			Pending	Excess to shortfall
B10	B07	S0190			Pending	Excess to shortfall
B16	B17	S0134			Pending	Excess to shortfall
B19	B09	S0134			Pending	Excess to shortfall
B22	B23	S0209			Pending	Excess to shortfall
B09	B19	S0285			Pending	Excess to shortfall

1. The plan in one sentence
2. Where stock flows
3. Approve or reject; logged

Live demo (view-only): stockflow-six-mu.vercel.app

Next: a 12-week pilot at 2–3 branches on real ERP and warehouse data

Weeks 1–4

Connect data

7 standard extracts, validated on load

Weeks 5–10

Shadow pilot

2–3 branches; planners approve or reject

Weeks 11–12

Decide

Monitor error and bias; retrain on drift

Limitations

- Synthetic data so far
- Plans one replenishment cycle
- Normal error model for safety stock
- Costs need finance approval

Right product, right branch: surgeons don't wait, so patients don't either.